TECHNICAL DEEP DIVE

Introduction to DBMS

Transforming raw data into organized intelligence for modern enterprises.

Defining the Foundation

Data

Raw, unorganized facts and figures. Can exist in raw form (unorganized) or processed form (meaningful).

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Example: Numbers "101, 102, 103" (Raw)
```

Database

A structured collection of data designed for efficient storage, retrieval, and manipulation by multiple users.

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Example: A table mapping "ID 101" to "John Doe" (Organized)
```

Where Databases Live

Banking

Managing customer transactions and accounts.

Airlines

Reservations and scheduling information.

Universities

Student registration and grades.

Telecom

Managing call records and billing.

Database Applications Diagram

Why we moved away from File Systems

- **Data Redundancy:** Duplicated data leading to inconsistency.
- **Isolation:** Data scattered in different files/formats.
- **Integrity:** Hard to enforce consistency constraints.
- **Security:** Difficult to provide restricted access.

The Major Risks

Atomicity problems (partial updates) and the lack of a robust Backup/Recovery mechanism made traditional file systems dangerous for critical enterprise data.

Two-Tier Architecture

Client-Server Model

The application resides on the client machine and communicates **directly** with the database server.

- Fast communication for local setups.
- Lower security due to direct DB access.

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Example: Railway Reservation (Local), MS-Access
Contact Manager.
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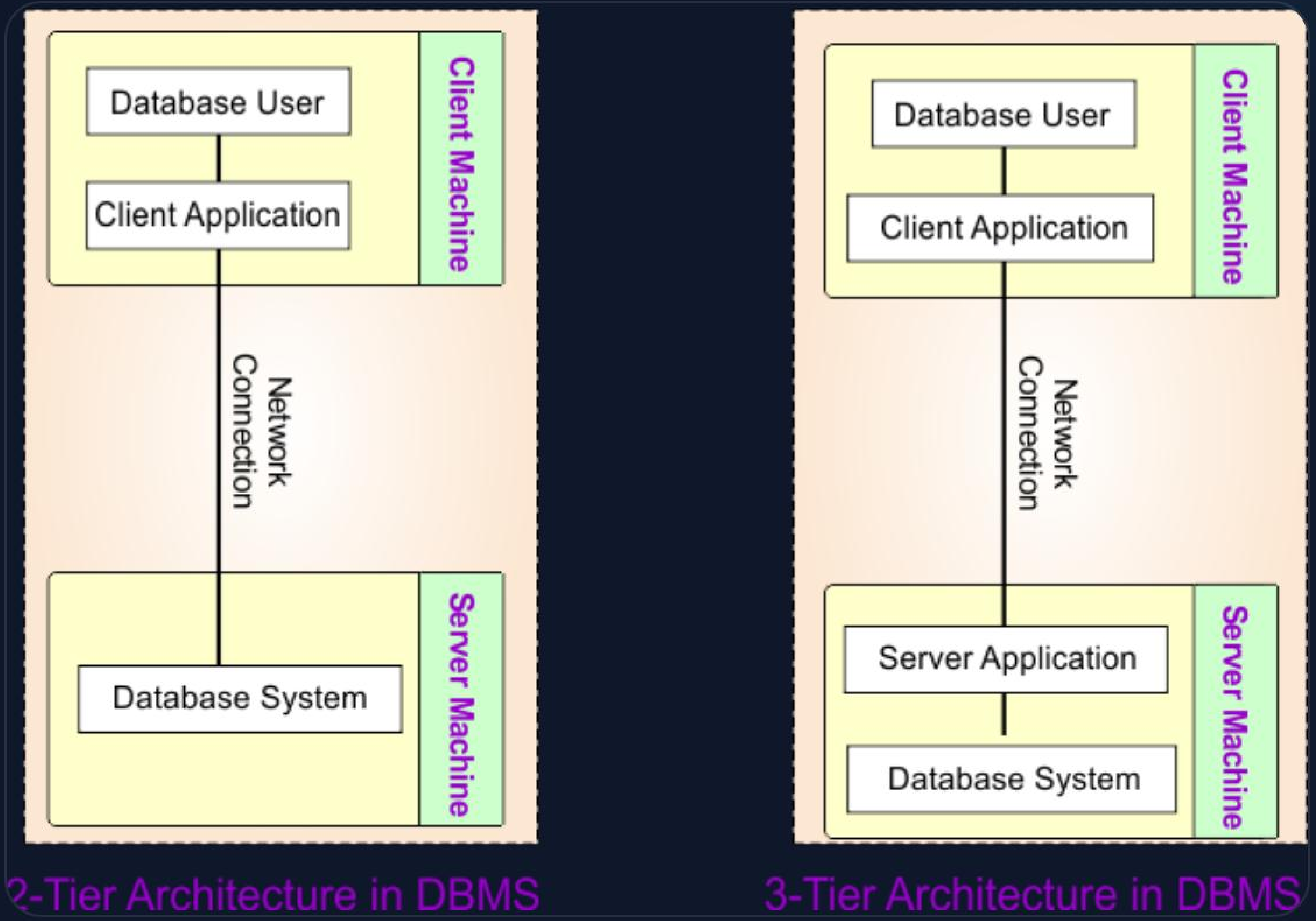


Three-Tier Architecture

Modern Web Architecture

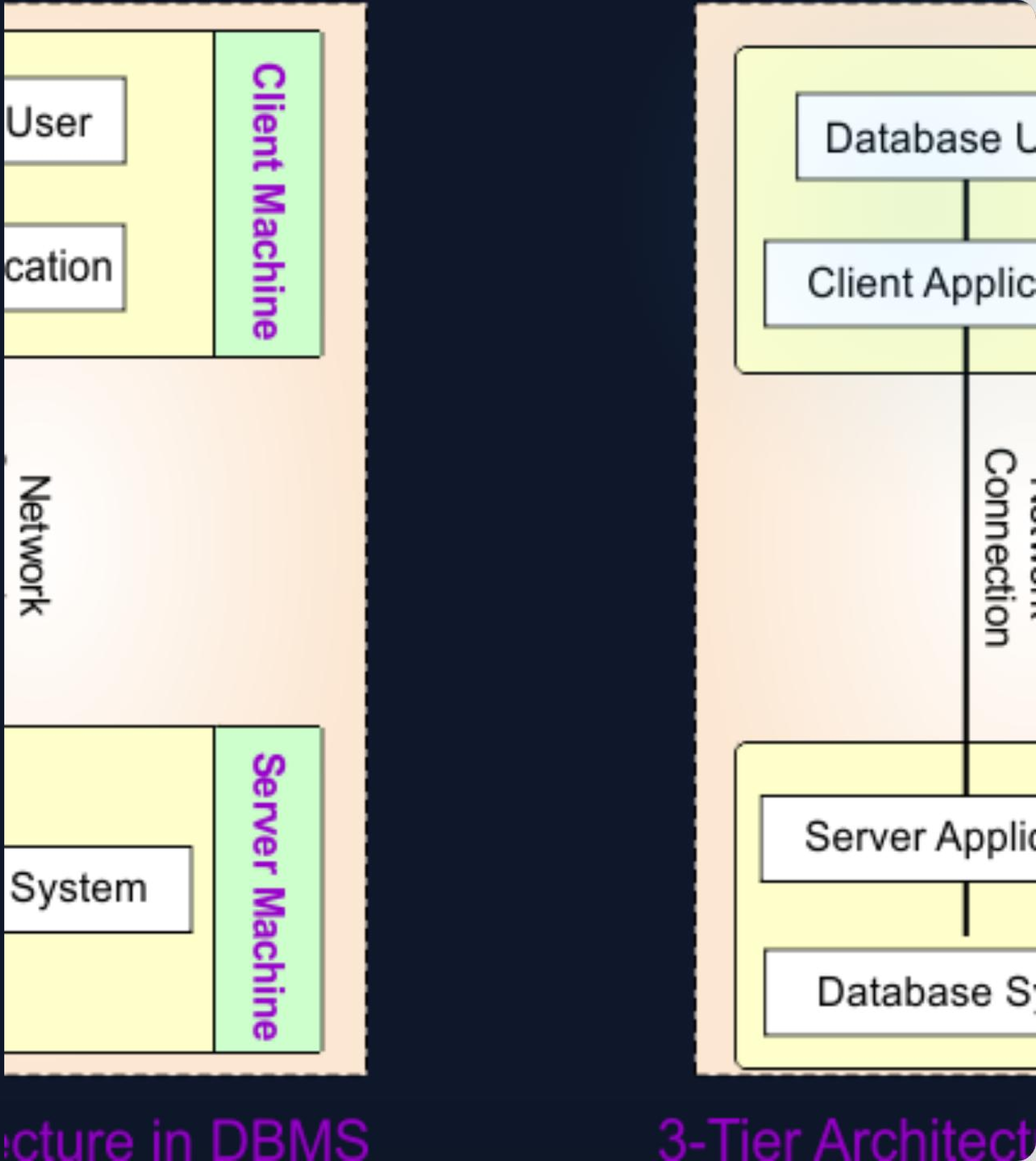
Separates the Presentation, Business, and Data layers for maximum security and scalability.

Presentation: UI layer (Browser/App).

Application: Logic & Middleware.

Database: Secure data storage.

`Example: Large E-commerce sites like Amazon.`



The Three Schema Architecture

Three Schema Diagram

1. External (View)

How users see the data. Multiple custom views possible.

2. Conceptual (Logical)

Defines entities, relationships, and constraints.

3. Internal (Physical)

How data is physically stored on the disk.

Ensuring Data Independence

The ability to change one layer without affecting the layers above it.

Type	Definition	Example Change
Physical Independence	Internal schema changes do not affect Conceptual level.	Moving from HDD to SSD or changing file structure.
Logical Independence	Conceptual schema changes do not affect External views.	Adding new attributes to a table.

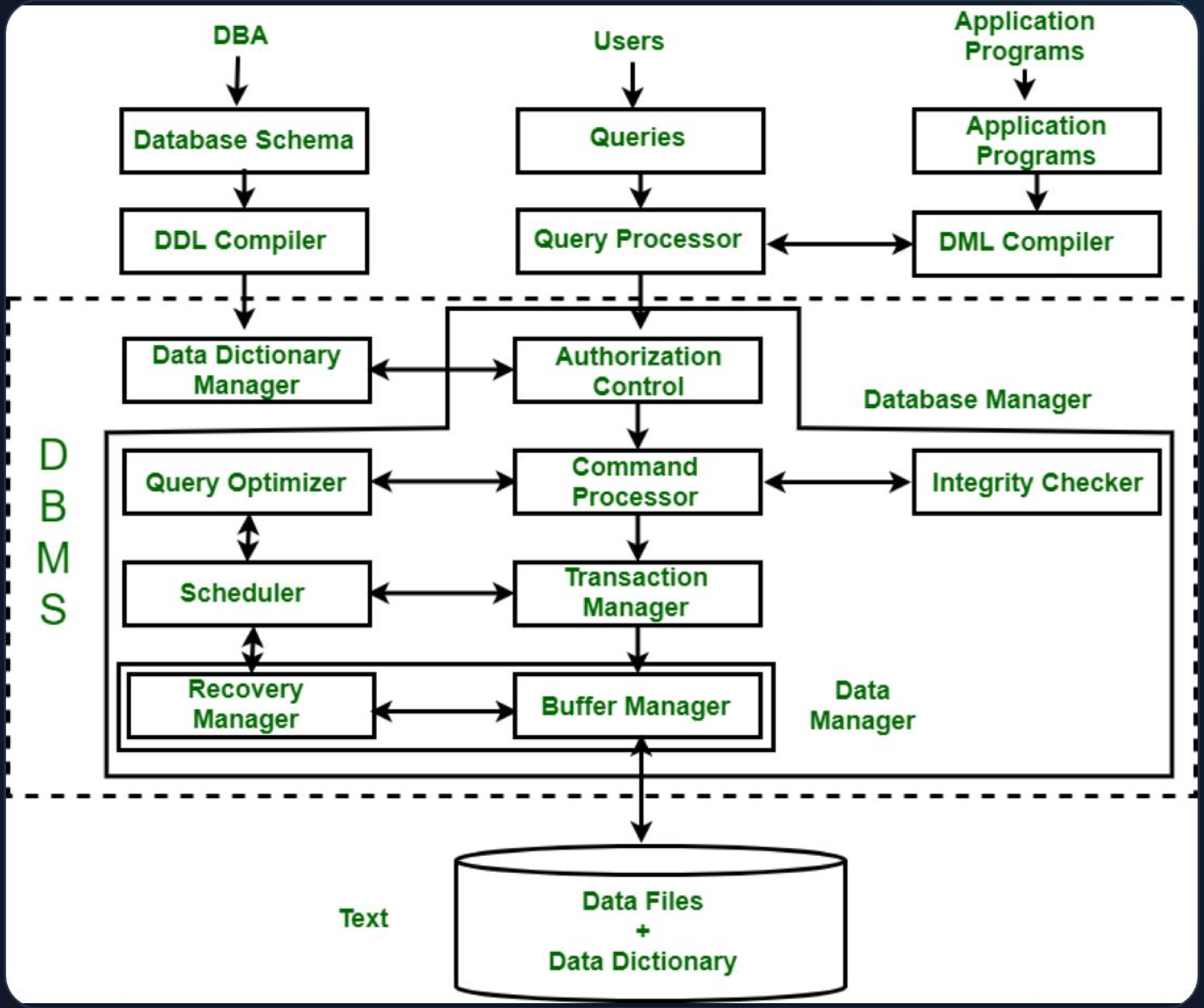
Inside the DBMS Engine

Storage Manager

Handles authorization, transactions, buffers, and file allocation on the disk.

Query Processor

Interprets DDL, compiles DML, and runs the query evaluation engine.



Who Uses the DBMS?

DBA
The super-users who define schema and control security/access.

Programmers
Write code that interacts with the database via applications.

Sophisticated
Analysts who write complex SQL queries directly.

Naive Users
Interact via pre-developed apps (e.g., bank tellers).

Designers
Responsible for the actual logical structure of data.

Casual Users
Occasional access to gather specific reports.

The Languages of Data

Language Type	Core Focus	Common Commands
DDL (Definition)	Schema/Structure Structure	`CREATE, ALTER, DROP, TRUNCATE`
DML (Manipulation)	Actual Data Records	`SELECT, INSERT, UPDATE, DELETE`
TCL (Control)	Transactions/Reliability	`COMMIT, ROLLBACK`

Metadata (Data about data) is stored in the **Data Dictionary** after DDL execution.

Modern Data Models

Relational

Rows and columns. The most standard model.

Object-Oriented

Stores data as Objects (e.g., MongoDB).

Hierarchical

Tree-like structure (Parent-Child).

ER Model

Pictorial blueprint using entities/links.

Data Models Comparison

Pro-Tip: Use 3-Tier and Relational models for most enterprise web applications!

Image Sources

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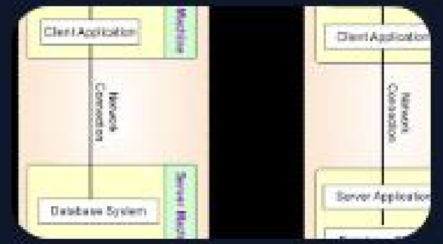

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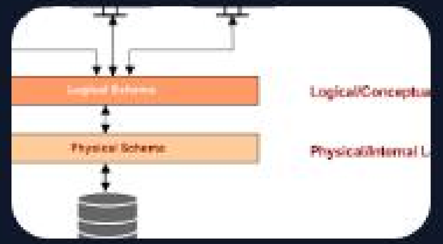

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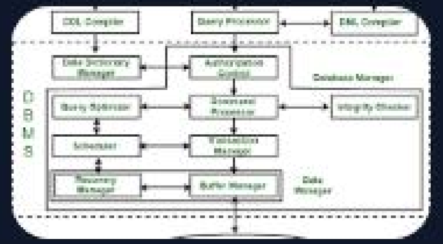

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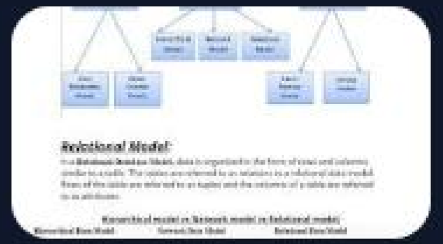

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